

From 3SAT to zero-finding with constant Jacobian

For every 3CNF formula ϕ , construct $K_\phi : \mathbb{R}^{3n+3m} \rightarrow \mathbb{R}^{3n+3m}$ such that

$$K_\phi^{-1}(0) \neq \emptyset \iff \phi \in \text{SAT},$$

$$\det JK_\phi \equiv (-3456)^{n+m}, \quad \deg K_\phi \leq 7.$$

1. Encode one bit

Fix $H : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with

$$\det JH \equiv -3456,$$

$$H^{-1}(0) = \{r_0, r_+, r_-\}.$$

Decode with $\beta(x, y, z) = x^2$:

$$\beta(r_0) = 0, \quad \beta(r_\pm) = 1.$$

2. Turn truth into a zero

Choose $\Delta \notin \text{im}(H)$ and define

$$T_s(v) = H(v) - (1 - s)\Delta.$$

For $s \in \{0, 1\}$,

$$\exists v : T_s(v) = 0 \iff s = 1.$$

Set $s = \text{"this clause is satisfied."}$

3. Stack the gadgets

Stack one variable block and one clause block per gadget:

$$K_\phi = (H(w_1), \dots, T_{s_m}(v_m)).$$

The Jacobian is block lower triangular. Hence

$$\det JK_\phi = (-3456)^{n+m}.$$